

Article

Dual-Detector Vision and Depth-Aware Back-Projection for Accurate Apple Detection and 3D Localisation for Robotic Harvesting

Tagor Hossain ^{1,*} , Peng Shi ¹  and Levente Kovacs ² ¹ School of Electrical and Mechanical Engineering, Adelaide University, North Terrace Campus, Adelaide, SA 5005, Australia² Research and Innovation Center, Obuda University, 1034 Budapest, Hungary

* Correspondence: mdtagor.hossain@adelaide.edu.au

Abstract

Accurate apple detection and precise three-dimensional (3D) localisation are essential for autonomous robotic harvesting in orchard environments, where occlusion, illumination variation, depth noise, and the similar colour appearance of fruits and surrounding leaves present significant challenges. This paper proposes a dual-detector vision framework combined with depth-aware back-projection to achieve robust apple detection and metric 3D localisation in real time. The method integrates the complementary strengths of YOLOv8 and Mask R-CNN through confidence-weighted fusion of bounding boxes and pixel-wise union of segmentation masks, producing stabilised two-dimensional (2D) apple representations under visually ambiguous conditions. The fusion results are converted into dense 3D representations through depth-guided projection within the camera coordinate system representing the visible fruit surface. A depth-consistency weighting strategy assigns higher influence to depth-reliable pixels during centroid computation, thereby suppressing noisy or occluded depth measurements and improving the stability of 3D fruit centre estimation, while local intensity normalisation standardises neighbourhood-level pixel intensities to reduce the impact of shadows, highlights, and uneven lighting, enabling more consistent segmentation and detection across varying illumination conditions. Experimental results demonstrate an accuracy of 98.9%, an mAP of 94.2%, an F1-score of 93.3%, and a recall of 92.8%, while achieving real-time performance at 86.42 FPS, confirming the suitability of the proposed method for robotic harvesting in challenging orchard environments.

Keywords: RGB-D orchard sensing; dual-detector fruit perception; depth-aware apple localisation; 3D fruit localisation in orchards; vision-based robotic harvesting



Academic Editor: David Portugal

Received: 12 January 2026

Revised: 10 February 2026

Accepted: 17 February 2026

Published: 22 February 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Vision-based perception plays a critical role in the development of intelligent agricultural robots, particularly for tasks such as fruit detection, localisation, and yield estimation. Accurate perception in orchard environments remains challenging due to complex backgrounds, heavy foliage occlusion, varying illumination conditions, and dense fruit clustering [1]. Robust apple detection and three-dimensional (3D) localisation are therefore essential prerequisites for reliable robotic harvesting and autonomous orchard monitoring systems. Recent advances in deep learning have significantly improved two-dimensional (2D) fruit detection performance using convolutional neural networks (CNNs) [2]. Deep learning-based detectors such as YOLO [3], Faster R-CNN [4], and Mask R-CNN [5] have

demonstrated strong capability in recognising apples under orchard conditions. However, purely RGB-based methods are inherently limited in complex orchard scenes, where visual ambiguity caused by occlusion, overlapping fruits, and illumination variation leads to unstable detection and inaccurate localisation [6]. Incorporating depth information through RGB-D sensing has therefore attracted increasing attention, as it provides complementary geometric cues that are less sensitive to colour similarity and lighting changes [7]. Several studies have explored RGB-D perception for fruit detection and localisation by combining depth maps with RGB imagery [8,9]. These approaches demonstrate that depth filtering and geometric constraints can effectively suppress background clutter and improve localisation accuracy. Nevertheless, many existing methods rely on a single detector, which makes them vulnerable to false positives, missed detections, and poor instance separation in densely clustered or occluded scenes [10]. Moreover, errors in 2D detection directly propagate to the subsequent 3D localisation stage, limiting the robustness of downstream perception for robotic harvesting tasks. To address occlusion and incomplete depth observations, point-cloud-based reasoning and frustum-constrained processing have been investigated for apple localisation under severe leaf and fruit occlusion [11]. These methods improve localisation accuracy for partially visible fruits; however, they depend heavily on the quality of initial 2D detections and segmentation masks. As a result, tightly integrated frameworks that combine robust multi-model 2D perception with depth-consistent 3D localisation remain relatively underexplored in orchard environments.

In this paper, a novel RGB-D perception framework is proposed for robust apple detection and 3D localisation in complex orchard environments. The proposed method integrates a dual-detector 2D apple detection strategy based on YOLOv8 and Mask R-CNN with depth-aware masking and 3D back-projection. By fusing confidence-weighted bounding boxes and instance-level segmentation masks, the proposed method improves detection stability under leaf occlusion, inter-fruit occlusion, and dense fruit clustering. Subsequently, depth-consistent back-projection enables accurate fruit-level 3D localisation using robust centroid estimation. Extensive experiments are conducted using public benchmark datasets and real-world orchard data collected in Lenswood, South Australia. The results demonstrate that the proposed method achieves reliable real-time performance while maintaining high detection accuracy and precise 3D localisation across diverse environmental conditions. The proposed framework differs from existing apple detection and localisation approaches in several key aspects. The main contributions of this paper are summarised as follows:

1. A paired RGB-depth data augmentation strategy that applies coordinated geometric and photometric transformations to enhance robustness against viewpoint variation, scale changes, and illumination fluctuations commonly encountered in orchard environments.
2. A dual-detector apple perception framework that integrates YOLOv8 and Mask R-CNN through confidence-weighted bounding box fusion and pixel-wise mask union, achieving robust and stable two-dimensional detection under occlusion, dense clustering, and complex backgrounds.
3. A depth-aware back-projection scheme that converts fused two-dimensional detections into fruit-level three-dimensional point sets within the camera coordinate system for accurate metric localisation.
4. A depth-consistency weighting strategy for robust centroid estimation, which suppresses unreliable depth pixels and mitigates the influence of occlusions and depth noise on 3D centre computation.
5. A local intensity normalisation mechanism that reduces illumination-induced appearance variations and enhances segmentation and detection consistency in both high-light and low-light orchard scenes.

6. Extensive evaluation on public benchmark datasets and real-world orchard data demonstrating high detection accuracy, precise 3D localisation, and real-time performance on an embedded platform.

These design choices distinguish the proposed system from established methodologies and contribute to its improved accuracy and stability in complex orchard environments.

The remainder of this paper is organised as follows. Section 2 reviews related work on vision-based fruit detection and localisation. Section 3 presents the proposed dual-detector apple detection and depth-aware 3D localisation methodology. Section 4 reports the experimental results and provides a detailed discussion. Finally, Section 5 concludes the paper and outlines directions for future research.

2. Related Work

The growing limitations of RGB-only perception in orchard environments have prompted increased research attention toward improving robustness against foliage occlusion, background clutter, and visually ambiguous scenes. Initial deep learning-based approaches to fruit detection largely relied on convolutional neural networks to improve two-dimensional detection accuracy in outdoor imagery [12–14]. These methods achieved strong performance under favourable conditions where fruits were clearly visible. Recent studies have also explored YOLO-based perception frameworks for agricultural robotics, including generative AI-driven decision-making pipelines that integrate visual detection with execution planning for fruit picking, demonstrating the effectiveness of YOLO architectures as reliable front-end detectors in complex orchard environments [15]. However, their reliability decreased substantially in realistic orchard scenarios involving dense canopies, overlapping fruits, and pronounced illumination variation [16]. Under such conditions, dependence on colour and texture information alone frequently resulted in missed detections and unstable localisation, thereby encouraging later studies to investigate alternative feature representations and complementary sensing modalities to enhance detection robustness in complex orchard environments. To alleviate these challenges, instance-level segmentation techniques were introduced to provide finer spatial delineation of individual fruits. Compared with bounding-box-based detection, pixel-wise segmentation enables more accurate separation of adjacent apples and improves fruit counting and shape estimation in clustered scenes [17–19]. Nevertheless, segmentation accuracy remains sensitive to background similarity and partial occlusion, particularly when fruits share colour characteristics with surrounding foliage. In addition, the computational complexity of segmentation-based approaches often limits their suitability for real-time robotic applications [20,21]. The integration of depth information has therefore emerged as an important direction for enhancing orchard perception. RGB-D sensing provides explicit geometric cues that complement appearance-based features, enabling improved discrimination between fruits and background structures [22,23]. Several studies have combined RGB imagery with depth maps to suppress background clutter, filter irrelevant regions, and enhance spatial consistency in fruit detection and localisation [24]. These results demonstrate that depth cues can substantially improve robustness under challenging lighting conditions and partial occlusion. Beyond detection, accurate three-dimensional (3D) localisation is a critical requirement for robotic harvesting systems, where precise fruit positioning directly influences grasp planning and manipulation success. Existing approaches typically estimate fruit positions by back-projecting 2D detections into 3D space using camera calibration parameters and depth measurements [25–27]. Although effective in controlled scenarios, such methods are highly dependent on the reliability of upstream 2D detection outputs. Inaccuracies in bounding boxes or segmentation masks propagate directly to the 3D reconstruction stage, leading to unstable localisation in cluttered orchard environments.

To address occlusion and incomplete depth observations, point-cloud-based reasoning and frustum-constrained processing have been investigated for fruit localisation under dense canopy structures [11,28]. By restricting 3D reasoning to regions of interest defined by 2D detections, these methods improve localisation accuracy for partially visible fruits. In addition, geometry-aware three-dimensional point cloud learning strategies have been shown to further improve spatial reasoning for precise cutting-point and manipulation target detection in unstructured field environments, highlighting the importance of combining geometric constraints with deep perception for agricultural robotics [29]. However, their effectiveness remains limited by the quality of the initial detection results and the ability to reliably separate individual fruit instances in close proximity. More recently, multi-model and ensemble-based perception strategies have been explored to exploit complementary strengths of different deep learning architectures [30–32]. By combining detectors with distinct inductive biases, these approaches aim to improve detection stability across diverse orchard conditions. In agricultural applications, such strategies have demonstrated improved robustness in scenes involving inter-fruit occlusion and background clutter. Nevertheless, many existing fusion approaches operate at a coarse decision level and lack tight integration with depth-aware geometric reasoning. Despite these advances, unified RGB-D perception frameworks that jointly address robust 2D detection, reliable instance separation, and depth-consistent 3D localisation remain relatively underexplored. Most existing pipelines rely on a single detection backbone or treat depth processing as a post hoc refinement step, limiting their ability to fully exploit complementary appearance and geometric information [33–35]. Furthermore, few studies explicitly focus on stabilising detection outputs prior to 3D reconstruction, which is critical for maintaining localisation accuracy in dynamic orchard environments. Recent trends in agricultural robotics emphasise the need for tightly coupled perception pipelines that integrate multiple detection modalities with geometric reasoning in a unified framework [36–40]. Such integration is particularly important for handling simultaneous challenges including leaf-induced occlusion, fruit-to-fruit overlap, scale variation, and complex background clutter. These gaps in the existing literature motivate the development of robust RGB-D perception architectures that combine complementary 2D detectors with depth-aware fusion and localisation strategies, as pursued in this work.

3. Dual-Detector Apple Detection and Depth-Aware 3D Localisation

The proposed method adopts a unified RGB-D perception architecture that integrates data augmentation, dual-detector two-dimensional (2D) apple detection, and depth-aware three-dimensional (3D) localisation within a single processing pipeline. Synchronised RGB and depth images are first subjected to paired data augmentation, including geometric transformations (flipping and scaling) and photometric adjustments (brightness, contrast, and hue), which are applied consistently to preserve spatial correspondence and enhance robustness to real-world variability. The augmented inputs are then processed by a dual-detector 2D apple detection module that jointly employs YOLOv8 and Mask R-CNN to exploit their complementary strengths. YOLOv8 serves as the primary object localisation component, providing efficient and reliable 3D position proposals through multi-scale feature extraction enhanced by spatial-channel reallocation, residual feature aggregation, and multi-receptive-field context modelling, making it effective under varying illumination and partial occlusion. In parallel, Mask R-CNN performs instance-level segmentation using an instance feature extractor with a Feature Pyramid Network (FPN) and dense boundary encoding, producing precise pixel-wise apple masks that accurately delineate fruit boundaries, particularly in cases of overlap or dense clustering.

YOLOv8 and Mask R-CNN were selected because they exhibit complementary strengths that are particularly beneficial in complex orchard environments. YOLOv8 follows a single-stage detection paradigm with multi-scale feature aggregation, enabling rapid and reliable localisation of apple candidates across a wide range of object sizes and viewpoints. This makes YOLOv8 well suited for real-time operation and for handling scenes containing many fruits distributed over large image areas. In contrast, Mask R-CNN adopts a two-stage detection and segmentation strategy that emphasises region refinement and pixel-level mask prediction, producing accurate instance boundaries even when fruits overlap or are partially occluded by foliage. While Mask R-CNN is more computationally demanding, its precise segmentation capability provides detailed shape information that is difficult to obtain from bounding-box-based detectors alone. By combining YOLOv8 and Mask R-CNN, the proposed framework exploits the fast and stable localisation of YOLOv8 together with the high-quality boundary delineation of Mask R-CNN. This complementary behaviour is particularly advantageous in scenarios involving dense fruit clustering, inter-fruit occlusion, and background clutter, where relying on a single detector often leads to missed detections or inaccurate instance separation. The fusion of their outputs therefore results in more reliable two-dimensional apple representations, which subsequently improve the accuracy and robustness of depth-aware three-dimensional localisation.

The outputs of both detectors are integrated through a stabilised detection fusion strategy that refines bounding boxes using mask-derived spatial constraints and enforces confidence-weighted agreement between detectors, resulting in robust and consistent 2D apple detections. Before fusion, the confidence scores produced by YOLOv8 and Mask R-CNN are normalised to a common scale to ensure numerical comparability. Both detectors are trained on the same apple datasets using identical annotation protocols and consistent optimisation settings, which promotes similar confidence behaviour across diverse orchard scenes. Empirical evaluation shows that high-confidence predictions from both networks consistently correspond to correct detections, indicating that the confidence-weighted fusion scheme provides a reliable measure of detection reliability. These fused detections are subsequently projected into 3D space using a depth back-projection module, where pixel-level depth values within each detected fruit region are mapped to the camera coordinate frame to form fruit-level 3D point sets. Depth consistency filtering is applied to suppress noise and outliers, followed by robust centroid estimation to obtain accurate 3D apple coordinates. By tightly coupling appearance- dual-detector perception with depth-aware geometric reasoning, the proposed architecture achieves reliable apple detection and localisation under challenging orchard conditions, including occlusion, illumination variation, and dense fruit clustering. The complete processing pipeline of the proposed method, including RGB-depth augmentation, dual-detector two-dimensional perception, and depth-guided three-dimensional localisation, is shown in Figure 1.

The method is organised into seven stages: (i) data augmentation, (ii) dual-detector fusion for 2D boxes and masks, (iii) depth back-projection to obtain a 3D point cloud, (iv) robustness to occlusions and illumination variation, (v) apple size estimation in both pixel and metric units, (vi) reconstruction of a 3D localisation, and (vii) evaluation using standard detection metrics. The detailed formulation of each component in the proposed framework is described comprehensively using mathematical equations.

Deep-learning detectors require diverse and representative training data to generalise to the wide range of viewpoints, scales, and lighting conditions commonly observed in orchards. Because collecting such variation-rich datasets is costly and time-consuming, data augmentation becomes an essential component of the proposed pipeline. In this work, we apply a set of geometric and photometric transformations that simulate realistic

appearance changes while preserving the structural integrity of the fruit. The following equations formalise these transformations and introduce the associated notation.

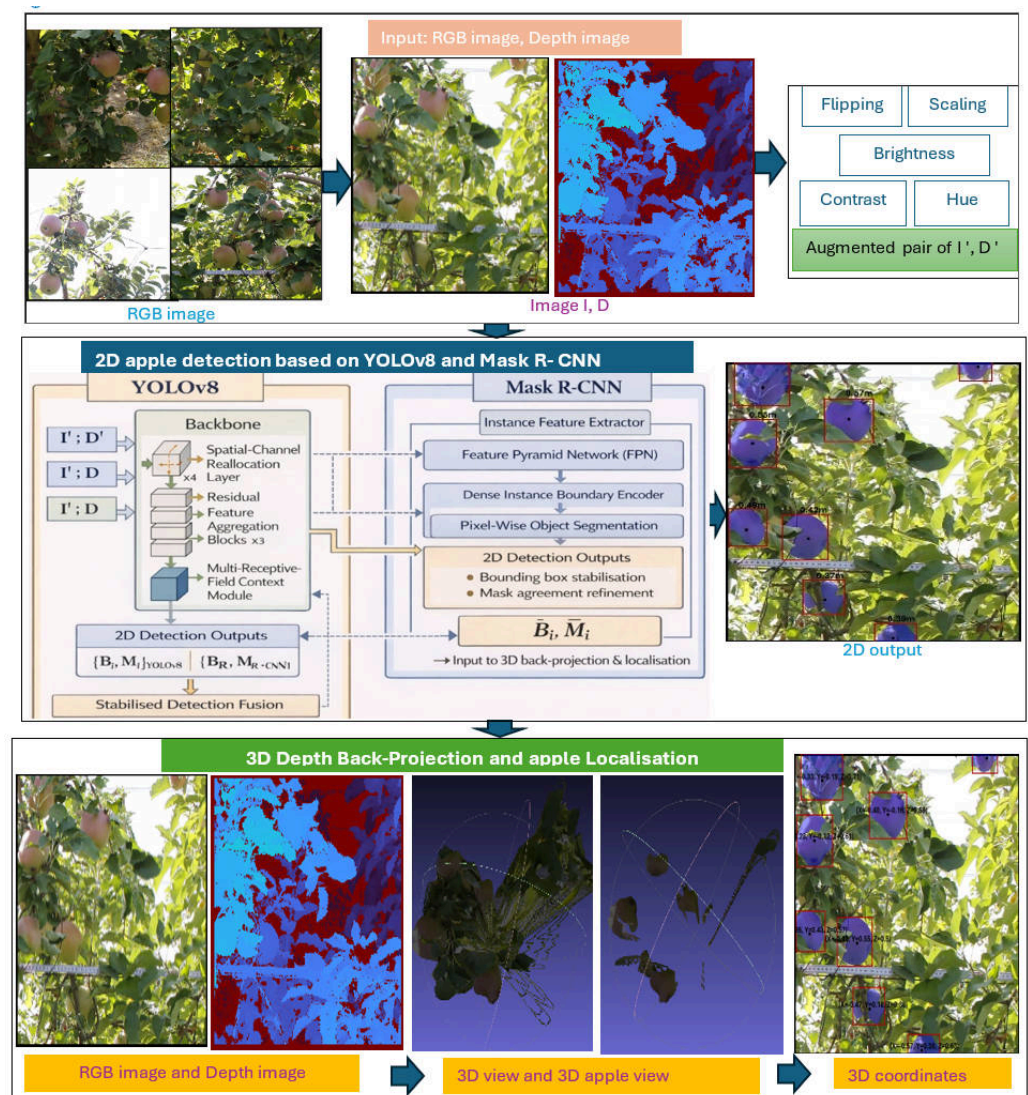


Figure 1. Overview of the proposed RGB-D perception architecture for apple detection and three-dimensional localisation. The framework integrates RGB-depth data augmentation, dual-detector two-dimensional apple perception using YOLOv8 and Mask R-CNN with stabilised fusion, and depth-aware back-projection for accurate fruit-level 3D position estimation.

The augmentation process begins with horizontal flipping, which mirrors the image across a vertical axis through its centre according to

$$I_{\text{flip}}(u, v) = I(W - u, v), \quad (1)$$

where W is the image width and $I(u, v)$ denotes the original pixel intensity. This ensures that the detector develops left-right invariance, allowing it to recognise apples regardless of whether they appear on the left or right side of the scene. To simulate variations in camera orientation, each image is additionally subjected to in-plane rotation, applied around the optical centre using

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} u - u_r \\ v - v_r \end{bmatrix} + \begin{bmatrix} u_r \\ v_r \end{bmatrix}, \quad (2)$$

where (u_r, v_r) is the rotation centre and θ is the rotation angle. This transformation encourages robustness to arbitrary viewpoint changes caused by uneven terrain or natural robot movement.

To account for the varying distances at which apples may appear, we apply anisotropic scaling, defined by

$$\begin{bmatrix} u_s \\ v_s \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}, \quad (3)$$

where s_x and s_y control the horizontal and vertical scaling. This operation adjusts the apparent size of the fruit, enabling the detector to learn scale-invariant features and maintain consistent performance across different viewing distances.

To model illumination changes commonly observed in orchards, a brightness adjustment is applied,

$$I_{\text{bright}}(u, v) = \gamma I(u, v), \quad (4)$$

where $\gamma > 0$ regulates global brightness. This reproduces conditions ranging from shaded canopy regions to strongly sunlit patches, reducing the likelihood of illumination-related detection errors. Beyond geometric transformation, orchard imagery also exhibits considerable photometric variation, so we further introduce a *contrast adjustment* to expand or compress the dynamic range of intensities,

$$I_{\text{contrast}}(u, v) = \lambda(I(u, v) - \mu) + \mu, \quad (5)$$

where λ is a contrast scaling factor and μ is the mean intensity.

The colour appearance of fruit is further diversified through a hue-shift transformation, applied in HSV space as

$$H'(u, v) = (H(u, v) + \Delta h) \bmod 360, \quad (6)$$

where Δh is the hue offset. This accounts for natural variations in apple colouration as well as lighting-induced colour shifts.

All geometric and photometric modifications are combined into a single augmentation pipeline, applied sequentially as

$$I' = T_{\text{hue}} \circ T_{\text{contrast}} \circ T_{\text{bright}} \circ T_{\text{scale}} \circ T_{\text{rotate}} \circ T_{\text{flip}}(I), \quad (7)$$

with corresponding transformations applied to the depth image to maintain pixel-wise alignment. This integrated process produces highly diverse yet physically consistent training samples, significantly enhancing detector robustness under the complex visual conditions typical of orchard environments.

Following augmentation, the enhanced image pair (I', D') is processed through two complementary detectors to obtain accurate 2D localisations of apples. The first detector, YOLOv8, produces a set of bounding boxes, instance masks, and confidence values according to

$$(B_Y, M_Y, c_Y) = \text{YOLOv8}(I', D'), \quad (8)$$

where B_Y denotes the predicted bounding boxes, M_Y the associated segmentation masks, and c_Y their confidence scores.

To refine these detections, the same input is passed through Mask R-CNN, which generates higher-precision boundaries via

$$(B_R, M_R, c_R) = \text{MaskRCNN}(I', D'), \quad (9)$$

where B_R , M_R , and c_R represent the bounding boxes, masks, and confidence values predicted by the second detector.

To combine the strengths of both detectors, the bounding boxes predicted by YOLOv8 and Mask R-CNN are fused using a confidence-weighted average,

$$B = \frac{c_Y B_Y + c_R B_R}{c_Y + c_R}, \quad (10)$$

where B_Y and B_R are the bounding boxes predicted by YOLOv8 and Mask R-CNN, respectively, and c_Y and c_R denote their associated confidence scores. This produces a single, stabilised bounding box B for each detected apple.

The segmentation masks predicted by the two detectors are combined through a pixelwise logical union,

$$M = 1[M_Y + M_R > 0], \quad (11)$$

where M_Y and M_R represent the masks generated by YOLOv8 and Mask R-CNN. The resulting mask M retains any pixel identified as fruit by either detector, providing a more complete segmentation of the apple. Although a pixelwise logical union of segmentation masks may potentially enlarge object regions when the two detectors partially disagree, this seemingly adverse effect is mitigated in the proposed framework through spatial and geometric constraints. Specifically, the union operation is applied only within the refined fused bounding box, and pixels that violate subsequent depth-consistency filtering are suppressed during 3D reconstruction. As a result, spurious regions introduced by partial detector disagreement are effectively removed in later processing stages. This design allows the mask union to improve fruit surface coverage under occlusion while avoiding uncontrolled mask expansion.

The combined detections for all apples in the image are then expressed compactly as

$$D = \{(B_i, M_i, c_i)\}_{i=1}^N, \quad (12)$$

where each tuple (B_i, M_i, c_i) corresponds to one detected fruit and includes its fused bounding box B_i , fused segmentation mask M_i , and the resulting confidence score c_i . The set D therefore contains the complete collection of detections for all N apples present in the augmented image and forms the basis for subsequent 3D reconstruction.

To estimate each apple's spatial location, the fused mask is projected into 3D using depth values from the corresponding pixels. Each pixel (u, v) is mapped into the camera coordinate frame via

$$q(u, v) = ((u - u_p) D(u, v) / f_u, (v - v_p) D(u, v) / f_v, D(u, v)), \quad (13)$$

where (u_p, v_p) is the principal point and f_u, f_v are the focal lengths. This transformation converts 2D image coordinates and their depths into 3D positions, providing the basis for reconstructing the visible portion of the fruit.

After back-projecting the pixels belonging to each fused mask, we obtain a discrete set of 3D points that represents the visible surface of the fruit. This set is defined as

$$Q = \{q(u, v) \mid (u, v) \in M\}, \quad (14)$$

where Q contains all 3D coordinates corresponding to pixels marked as fruit in the segmentation mask M . Aggregating these points forms a partial surface reconstruction that is robust to missing regions or incomplete views.

To estimate the fruit's spatial centre, we compute the geometric median of the 3D points in Q , expressed as

$$c = \arg \min_y \sum_{x \in Q} \|x - y\|_2, \quad (15)$$

where c is the 3D centroid and $\|\cdot\|_2$ denotes the Euclidean norm. The geometric median provides a stable estimate of the fruit centre because it reduces sensitivity to noisy depth measurements or stray pixels that may be incorrectly included in the mask.

Depth noise caused by foliage, branches, or abrupt shading transitions can distort the surface reconstruction. To mitigate such effects, we assign each pixel a depth-weight

$$w(p) = \exp\left(-\frac{(D(p) - z_{\text{avg}})^2}{2\sigma^2}\right), \quad (16)$$

where $p = (u, v)$ is a pixel, $D(p)$ denotes its depth value, z_{avg} is the mean depth of all mask pixels, and σ is a tolerance parameter. The parameter σ controls the sensitivity of the weighting function to depth deviations and therefore determines the strength of noise suppression. In this work, σ is selected empirically on preliminary experiments conducted on the validation set, where different values were evaluated to balance outlier rejection and preservation of valid fruit surface points. A small σ produces a narrow weighting distribution, aggressively down-weighting pixels whose depths deviate from the mean and effectively suppressing noise caused by foliage, branches, and mixed pixels, but may discard legitimate surface points. Conversely, a large σ results in smoother weighting, retaining more pixels at the risk of admitting noisy measurements. The chosen σ provides a compromise between these two extremes, yielding stable centroid estimates while maintaining sufficient surface coverage. This setting was found to be robust across different orchard scenes and illumination conditions. A Gaussian model is adopted for depth-consistency weighting because it provides a smooth and symmetric attenuation of depth outliers while preserving pixels close to the dominant fruit surface. Although a fixed σ is used in this work, its value is selected to be proportional to the typical depth variation observed within apple surfaces in the training data, which implicitly accommodates moderate differences in fruit size and viewing distance. Larger fruits or closer viewpoints naturally exhibit greater surface depth variation, whereas smaller or distant fruits produce tighter depth distributions; the selected σ was empirically chosen to remain effective across these cases. Validation experiments confirmed that the chosen setting yields stable centroid estimation under varying occlusion levels without requiring per-fruit parameter adaptation, offering a practical compromise between robustness and computational simplicity.

Using these depth-consistent weights, we refine the 3D centre estimate as

$$c_w = \frac{\sum_{p \in M} w(p) q(p)}{\sum_{p \in M} w(p)}, \quad (17)$$

where $q(p)$ is the 3D back-projected point for pixel p . This weighted centre places greater emphasis on depth-reliable pixels and thus improves robustness when parts of the fruit are partially hidden behind leaves or branches.

To compensate for illumination variations across the fruit surface, we apply local intensity normalisation, defined by

$$I_n(u, v) = \frac{I'(u, v) - \mu_L}{\sigma_L}, \quad (18)$$

The neighbourhood size used to compute the local mean μ_L and standard deviation σ_L is defined by a square window centred at each pixel, where $I'(u, v)$ is the augmented image

and the statistics are calculated within a neighbourhood around pixel (u, v) . In the current implementation, a fixed $k \times k$ window with $k = 9$ is used for all experiments. This size was selected empirically through validation to provide a good balance between local contrast enhancement and noise robustness. Smaller windows tend to amplify sensor noise and local texture fluctuations, whereas larger windows oversmooth local details and reduce the effectiveness of normalisation near fruit boundaries. The chosen window size therefore preserves local structural details while sufficiently compensating for gradual illumination changes. In addition, because μ_L and σ_L are computed independently for each pixel on its surrounding neighbourhood, the normalisation operation adapts automatically to spatially varying lighting conditions across the image, enabling consistent enhancement under both strong illumination and low-light scenarios.

After normalising local pixel intensities, we estimate the apparent fruit size directly from the segmentation mask. Assuming a near-circular silhouette, the effective pixel diameter is approximated by

$$d_{\text{pix}} = \sqrt{\frac{4|M|}{\pi}}, \quad (19)$$

where $|M|$ denotes the number of mask pixels and $\pi \approx 3.1416$. This relation follows from the area-to-diameter formula for a circle and provides a convenient approximation of the fruit's diameter in pixel units without requiring explicit contour fitting. The metric diameter estimation in Equation (19) assumes a near-circular silhouette as a simplifying approximation of the apple's projected shape. While real apples may exhibit slight elongation and segmentation masks may be incomplete under partial occlusion, this assumption is used only to obtain an initial size estimate rather than a precise shape model. In practice, the influence of non-circular or partially visible silhouettes is reduced by averaging depth over all mask pixels and by subsequent geometric analysis of lateral and depth extents (Equations (23) and (24)), which account for ellipsoidal shape variations and perspective effects. Empirical evaluation indicates that the resulting metric diameter remains stable across different fruit orientations and moderate occlusion levels, making the approximation suitable for real-time harvesting applications where computational efficiency is required.

To account for perspective effects, we compute the average depth across all mask pixels as

$$z_{\text{avg}} = \frac{1}{|M|} \sum_{p \in M} D(p), \quad (20)$$

where $D(p)$ is the depth at pixel p . Averaging the depth over the entire mask yields a stable estimate of the fruit's distance from the camera by reducing the influence of local curvature, noise, or partially occluded regions.

Combining the apparent pixel diameter with the estimated depth, we convert the fruit size into metric units using the pinhole camera model,

$$d_m \approx d_{\text{pix}} \cdot \frac{z_{\text{avg}}}{f_u}, \quad (21)$$

where d_m is the diameter in metres and f_u is the horizontal focal length expressed in pixels. This linear scaling arises from the geometric projection relation, enabling an efficient metric diameter estimate without performing full 3D sphere fitting.

To maintain consistency between image and world coordinates, we use the camera projection matrix to map 3D points into pixel coordinates,

$$Z \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_u & 0 & u_p \\ 0 & f_v & v_p \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}, \quad (22)$$

where (X, Y, Z) are the 3D camera-frame coordinates, (u, v) are the corresponding pixel coordinates, f_u and f_v are the horizontal and vertical focal lengths, and (u_p, v_p) is the principal point. This compact form expresses the standard pinhole projection, ensuring geometric consistency between the fruit's 3D structure and its 2D image footprint.

Finally, we compute the fruit's lateral 3D extent from the extreme points of the reconstructed surface,

$$d_x = X_R - X_L, \quad d_y = Y_B - Y_T, \quad (23)$$

where X_R and X_L are the rightmost and leftmost 3D coordinates along the X -axis, and Y_B and Y_T are the bottom and top coordinates along the Y -axis. These spans capture the horizontal and vertical extents of the fruit in metric space, providing further geometric cues for size estimation and shape analysis.

To estimate the depth extent of the fruit from its lateral spans, we fit a quadratic regression model that relates the observed horizontal and vertical extents to the underlying 3D depth of the fruit. The estimated depth is expressed as

$$d_z \approx a_0 + a_1 d_x^2 + a_2 d_y^2 + a_3 d_x + a_4 d_y, \quad (24)$$

where d_z denotes the fruit's depth extent, d_x and d_y are the lateral spans defined in Equation (23), and $a_0, \dots, a_4$ are regression coefficients learned from calibration fruit. The regression coefficients are obtained using a supervised calibration procedure in which a set of representative apples with measured metric dimensions is used to fit the quadratic model via least-squares regression. The calibration set includes fruits spanning different sizes, shapes, and maturity stages, ensuring coverage of typical intra-class variability. Because the regression relates geometric extents rather than appearance features, the learned mapping captures structural properties that are largely independent of colour or cultivar. Validation experiments on both benchmark datasets and real orchard data indicate that the calibrated model provides stable depth-extent estimates across different apple varieties and moderate shape variations, supporting its generalisability in practical orchard environments. This quadratic formulation captures the empirical relationship between lateral and depth extents, accommodating ellipsoidal shape variations and perspective foreshortening that occur at different viewing angles.

With the lateral and depth spans determined, we compute the Euclidean distance from the camera to the fruit centre as

$$D = \sqrt{X_c^2 + Y_c^2 + Z_c^2}, \quad (25)$$

where (X_c, Y_c, Z_c) are the 3D coordinates of the fruit centre. This range estimate is useful for robot reach planning, collision avoidance, and determining safe approach distances for manipulation. Finally, each 3D vertex is projected back into pixel space using the camera model of Equation (22). This enables consistent visualisation and alignment between the reconstructed 3D fruit and the original image data, facilitating quality inspection, annotation, or multi-view fusion.

To quantify detection performance, we report standard evaluation metrics commonly used in multi-fruit localisation tasks. Precision is defined as

$$P = \frac{TP}{TP + FP}, \quad (26)$$

where TP denotes the true positive detections number and FP shows false positives. Precision measures the proportion of system detections that correctly correspond to apples, penalising erroneous predictions caused by leaves, branches, or background clutter.

Complementing precision, recall evaluates the system's capability to identify all apples present in the scene, and is given by

$$R = \frac{TP}{TP + FN}, \quad (27)$$

where FN corresponds to false negatives, i.e., apples present in the image but missed by the detector. Recall therefore captures completeness, rewarding systems that successfully identify dim, occluded, or partially visible fruit instances.

To provide a balanced measure that incorporates both precision and recall, we compute the harmonic mean of the two metrics, known as the F_1 -score:

$$F_1 = \frac{2PR}{P + R}. \quad (28)$$

The F_1 -score condenses performance into a single interpretable value, making it suitable for comparing different detection models or evaluating the system under varying levels of occlusion and scene complexity. The complete procedure of the proposed apple detection and three-dimensional localisation method is summarised in Algorithm 1.

The complete procedure for our apple detection and 3D localisation pipeline is summarised in Algorithm, which outlines the augmentation steps, dual-detector fusion, depth projection, and geometric reconstruction stages used to obtain metric fruit dimensions.

Algorithm 1: Apple Detection & 3D Localisation (Proposed).

Input: I, D , intrinsics $K = (f_u, f_v, u_c, v_c)$, detectors YOLOv8, Mask R-CNN, coeffs $\alpha_0 \dots \alpha_4$

Output: $\mathcal{D}$ and optionally P, R, F_1

(Training only) Augment (I, D) with flip/rotate/scale/photometric ops (Equations (1)–(7)); denote (I', D') .

(Inference) set $(I', D') \leftarrow (I, D)$. (Equations (1)–(7))

// 2D detection and mask fusion

$(B_Y, M_Y, c_Y) \leftarrow \text{YOLOv8}(I', D')$; $(B_R, M_R, c_R) \leftarrow \text{MaskRCNN}(I', D')$

$B \leftarrow \frac{c_Y B_Y + c_R B_R}{c_Y + c_R}$ (Equation (8))

$M \leftarrow \mathbf{1}[M_Y + M_R > 0]$ (Equation (9))

$c \leftarrow \max(c_Y, c_R)$ (Equation (10))

Extract $\{(B_i, M_i, c_i)\}_{i=1}^N$ (Equations (11) and (12)).

Algorithm 1: *Cont.*

```

for  $i = 1$  to  $N$  do
    // Back-project mask to 3D
     $q(u, v) = ((u - u_c)D / f_u, (v - v_c)D / f_v, D)$  (Equation (13))
     $Q_i \leftarrow \{q(u, v) \mid (u, v) \in M_i\}$  (Equation (14))
    Projection constraint check (Equation (15))
    // Depth consistency weights
     $z_{\text{avg}} = \frac{1}{|M_i|} \sum_{p \in M_i} D(p)$  (Equation (16))
     $w(p) = \exp\left(-\frac{(D(p) - z_{\text{avg}})^2}{2\sigma^2}\right)$  (Equation (17))
    // Weighted centroid
     $\mathbf{c}_i = \frac{\sum_{p \in M_i} w(p) q(p)}{\sum_{p \in M_i} w(p)}$  (Equation (18))
    // Size estimates
     $d_{\text{pix}} = \sqrt{4|M_i|/\pi}$  (Equation (19))
     $d_{m,i} = d_{\text{pix}} \cdot z_{\text{avg}} / f_u$  (Equation (20))
     $(d_{x,i}, d_{y,i}) = \max_X(Q_i) - \min_X(Q_i), \max_Y(Q_i) - \min_Y(Q_i)$  (Equation (21))
    // Depth extent regression
     $d_{z,i} = \alpha_0 + \alpha_1 d_{x,i}^2 + \alpha_2 d_{y,i}^2 + \alpha_3 d_{x,i} + \alpha_4 d_{y,i}$  (Equation (24))
     $D = \sqrt{X_c^2 + Y_c^2 + Z_c^2}$  (Equation (25))
    Store  $(B_i, M_i, c_i, \mathbf{c}_i, d_{m,i}, d_{x,i}, d_{y,i}, d_{z,i}, \{V_{i,j}\})$  in  $\mathcal{D}$ .

(Optional evaluation)
     $P = \frac{TP}{TP+FP}$  (Equation (26))
     $R = \frac{TP}{TP+FN}$  (Equation (27))
     $F_1 = \frac{2PR}{P+R}$  (Equation (28))
return  $\mathcal{D}$  and (optional)  $P, R, F_1$ 

```

4. Experimental Results and Discussion

4.1. Experimental Setup

The experimental setup of the proposed method is illustrated in Figure 2, which presents the embedded computing platform, the RGB-D sensing configuration, and the real-world orchard environment used for field trials. All experiments were conducted on an NVIDIA Jetson AGX- embedded system, shown in Figure 2a, which served as the primary on-board processing unit. The Jetson AGX integrates a multi-core ARM CPU and an NVIDIA GPU with dedicated CUDA and Tensor cores, supported by 32 GB of LPDDR4x VRAM. This configuration enables efficient execution of deep learning- detection networks and depth-aware three-dimensional perception tasks in real time. All runtime performance results, including the reported detection inference speed of 86.42 FPS, were measured on this embedded NVIDIA Jetson AGX platform using GPU-accelerated inference. The reported frame rate corresponds to the two-dimensional detection stage rather than the full end-to-end pipeline and reflects performance achieved with an optimised model configuration. This result demonstrates that the proposed detection framework is suitable for practical deployment on embedded robotic harvesting systems requiring low-latency perception. To support extended data acquisition and storage of RGB images, depth maps, and intermediate processing outputs, an external 500 GB storage device was connected to the platform.

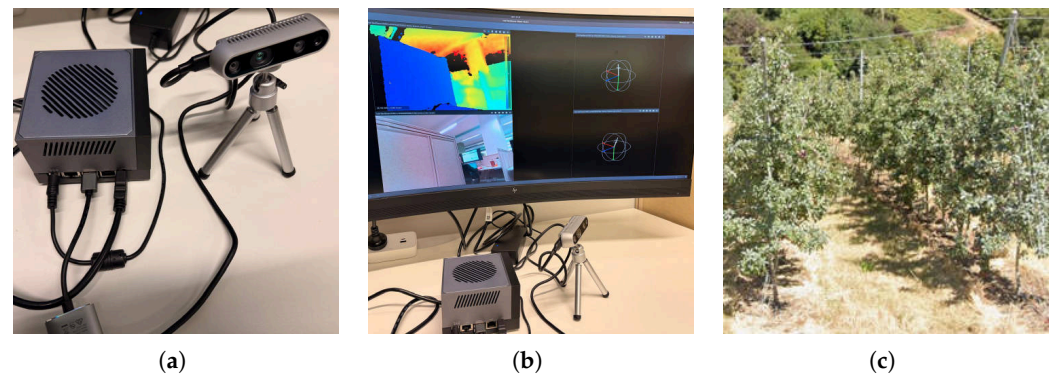


Figure 2. Experimental setup of the proposed method. (a) Embedded NVIDIA Jetson AGX computing platform used for on-board processing. (b) RGB-D sensing configuration showing the camera connected to the embedded platform. (c) Apple orchard in Lenswood, South Australia, where real-world field experiments were conducted.

As shown in Figure 2b, an RGB-D camera was interfaced with the Jetson AGX to acquire synchronised colour and depth streams. The camera provides aligned per-pixel depth measurements together with high-resolution RGB imagery, ensuring accurate correspondence between visual appearance and geometric information. This alignment is essential for the proposed method, which relies on depth masking, geometric back-projection, and fruit-level three-dimensional localisation. During data collection, the RGB-D camera was operated in a handheld manner to emulate realistic sensing conditions encountered in orchard environments. All perception, detection, and localisation processes were executed directly on the embedded platform, demonstrating the feasibility of deploying the proposed method for real-time, on-device operation. For network training and validation, the apple dataset was divided into 2290 training images and 801 validation images. To maintain consistency across experiments and optimise computational efficiency, all input images were resized to a fixed resolution of 896×896 pixels during training. Two complementary datasets were employed to develop and evaluate the segmentation component of the proposed apple detection framework. The first dataset was the publicly available MinneApple dataset, which contains 1000 annotated orchard images and is widely used as a benchmark in fruit detection research. The second dataset was the Amodal Apple dataset, consisting of 3019 images, which provides a broader range of apple appearances captured under diverse orchard conditions. To further characterise the representativeness of the training data, the MinneApple and Amodal Apple datasets were analysed in terms of occlusion level, lighting condition, and apple appearance. The datasets contain samples spanning three approximate occlusion categories: low occlusion (fully visible fruits), medium occlusion (partial leaf or branch coverage), and high occlusion (severely occluded or overlapping fruits). Images also cover diverse illumination conditions, including strong sunlight, moderate daylight, shadow-dominated scenes, and low-light environments, reflecting realistic orchard variability. In addition, both datasets include multiple apple varieties with noticeable differences in colour, size, and surface texture, ranging from green to yellow and red cultivars, as well as varying maturity stages. This diversity ensures that the training data expose the proposed model to a wide range of visual and geometric variations, supporting robust generalisation across different orchard scenarios. The combined use of these datasets exposes the model to variations in occlusion, illumination, scale, and fruit arrangement, thereby improving generalisation performance. In addition to dataset-based evaluation, real-world field experiments were conducted in an apple orchard located in Lenswood, South Australia, as shown in Figure 2c. During these trials, RGB-D data were collected by walking through the orchard while holding the RGB-D camera by

hand. This data acquisition strategy reflects realistic operating conditions for mobile robotic harvesting systems, including natural illumination changes, viewpoint variation, and complex background clutter. Together, the controlled dataset experiments and the field trials provide a comprehensive evaluation of the proposed method under both experimental and real-world orchard conditions. All real-world orchard experiments were conducted in Lenswood, South Australia, under naturally varying outdoor conditions. Data were collected across different times of day, including morning and afternoon periods, resulting in diverse illumination levels ranging from soft diffuse lighting to strong direct sunlight. Experiments were performed under mostly clear and partly cloudy weather, with occasional light wind, reflecting typical orchard operating conditions. The evaluated trees exhibit common open-canopy and semi-dense architectures with varying foliage density, branch arrangements, and fruit clustering patterns. These variations ensure that the collected data are representative of practical orchard environments and facilitate reproducibility of the experimental conditions.

4.2. Results and Discussion

Fruit detection and 3D localisation performance of the proposed framework is illustrated in Figures 3 and 4 under representative challenging orchard conditions. In Figure 3, first row, apples are partially occluded by surrounding foliage, where the RGB image alone provides weak visual contrast between fruit surfaces and background leaves, making reliable detection difficult. In this case, the depth mask highlights foreground structures and facilitates effective level separation, which suppresses irrelevant foliage regions while retaining apple candidates. As a result, the segmentation stage preserves coherent fruit boundaries, enabling stable and accurate two-dimensional (2D) detection despite significant leaf-induced occlusion. The corresponding three-dimensional (3D) processing results shown in Figure 4, first row, further demonstrate that the retained depth information supports consistent fruit-level point cloud reconstruction, allowing accurate spatial localisation even when only partial fruit surfaces are visible.

In Figure 3, second row, the scene contains apple-to-apple occlusion, where neighbouring fruits overlap in the image plane and produce ambiguous visual boundaries. Under such conditions, the proposed dual-detector fusion benefits from complementary bounding-box cues and instance-level segmentation, improving separation between adjacent apples. Depth-aware masking further refines the segmentation by removing background interference, resulting in reliable 2D detections for individual fruits. This improved instance separation is reflected in Figure 4, second row, where the reconstructed 3D views show clearly separated fruit point sets and stable localisation outputs, indicating robustness to inter-fruit occlusion.

The third row of Figure 3 illustrates scenarios involving densely clustered apples distributed along branches, where minimal spacing between fruits often leads to merged detections in conventional approaches. Despite these challenging conditions, the proposed framework successfully maintains instance-level segmentation and consistent 2D detection for multiple closely spaced apples. As shown in Figure 4, third row, the depth-guided back-projection reconstructs distinct fruit-level 3D representations, enabling precise localisation of each apple within the cluster. Overall, these qualitative results confirm that the proposed method effectively integrates appearance-based perception with geometric depth reasoning to handle foliage occlusion, inter-fruit overlap, and dense clustering in realistic orchard environments.

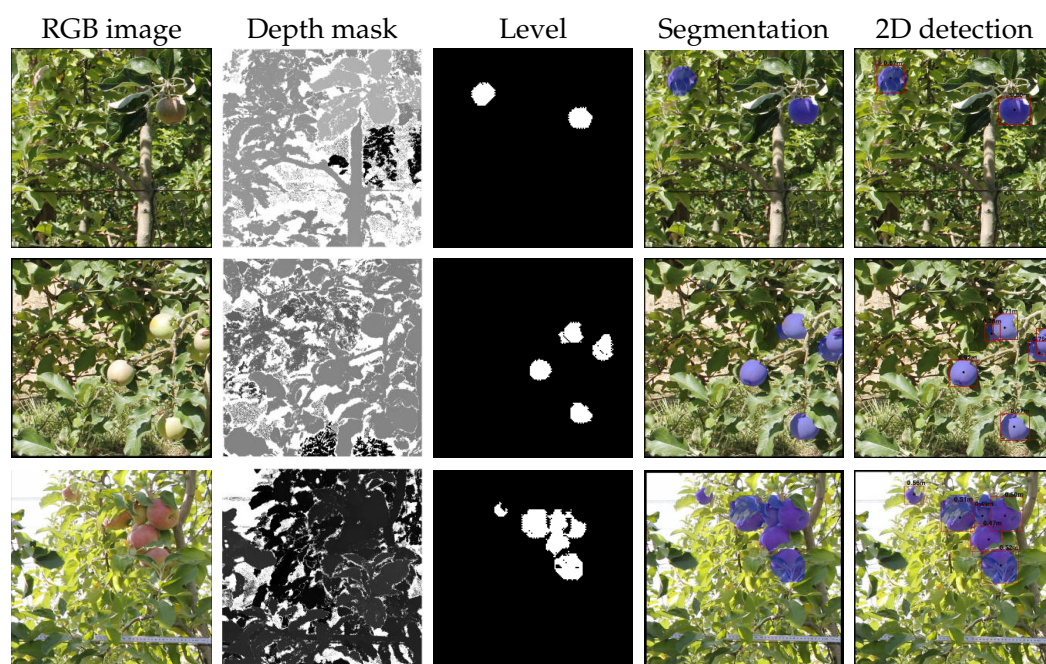


Figure 3. Qualitative assessment of the proposed 2D apple detection pipeline under complex orchard scenarios. Each row illustrates sequential processing stages, including the original RGB image, depth-derived masking, level-based separation, segmentation output, and the final 2D detection result. The red bounding boxes in the last column indicate the detected apple instances in the image plane. The first row presents results under leaf-induced occlusion, the second row demonstrates performance in the presence of inter-fruit occlusion, and the third row highlights reliable detection of densely grouped apples along tree branches.

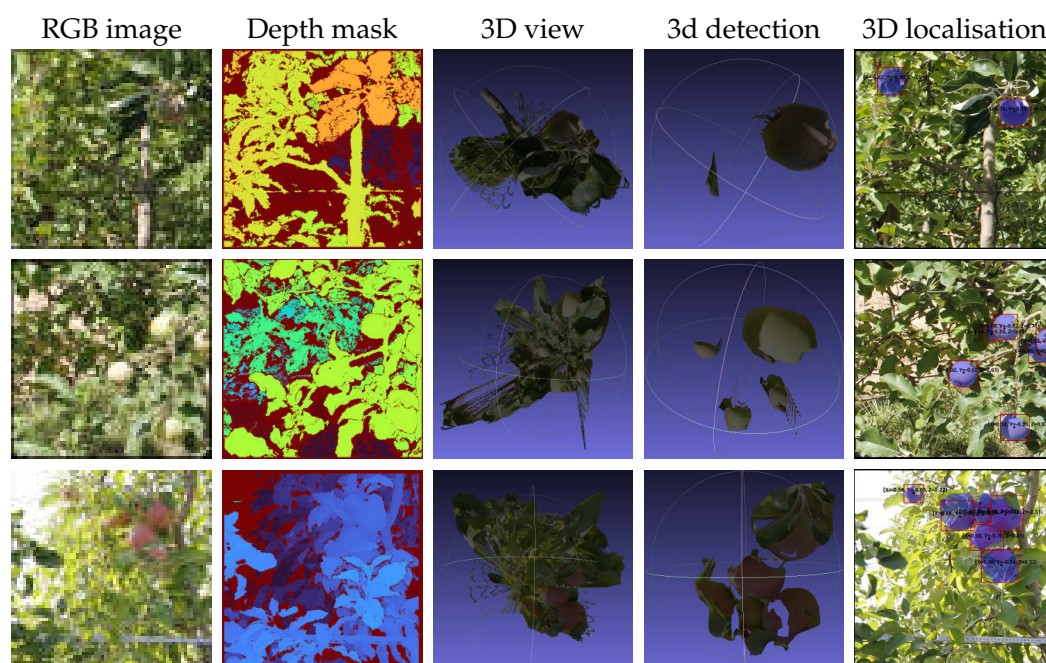


Figure 4. Qualitative evaluation of the proposed three-dimensional apple localisation framework in complex orchard environments. Each row presents consecutive stages of the processing pipeline, including the RGB input image, depth-based masking result, reconstructed 3D scene representation, fruit-level 3D detection, and the final localisation output. The red bounding boxes shown in the last column indicate the detected apples and their corresponding localisation results projected onto the image plane. The first row demonstrates robustness to foliage-induced occlusion, the second row illustrates performance under inter-fruit occlusion, and the third row highlights accurate localisation of densely clustered apples along branches.

The qualitative performance of the proposed framework under challenging illumination and appearance conditions is showing in Figures 5 and 6. These results complement the occlusion-focused analysis by examining the robustness of the proposed method to strong lighting variation, low illumination, and colour ambiguity, which commonly degrade vision-based perception in orchard environments.

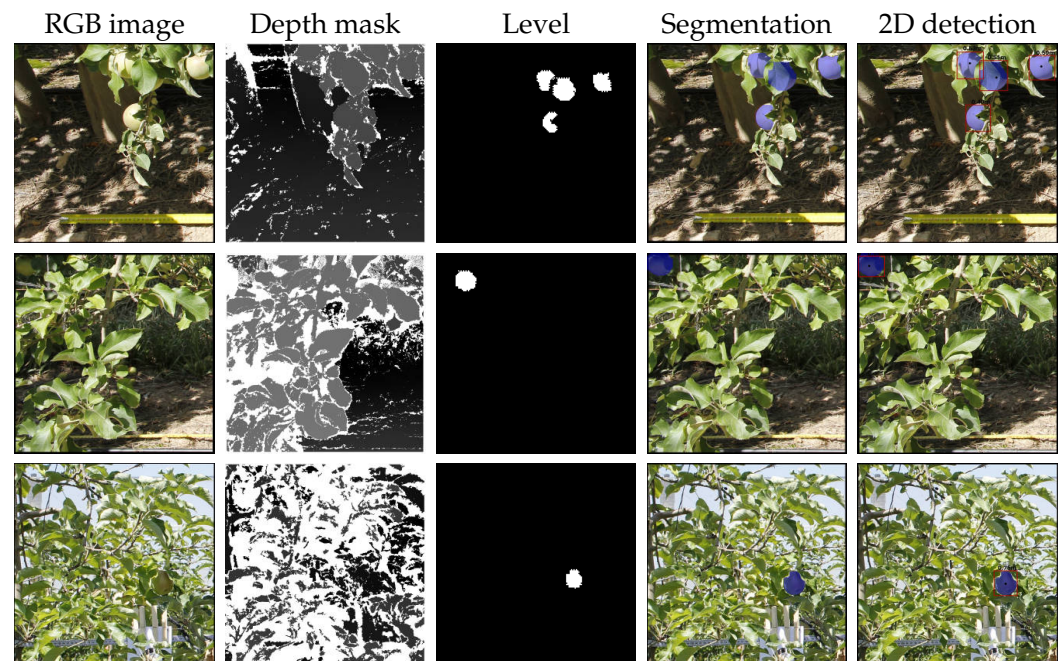


Figure 5. The first row demonstrates effective apple detection under strong illumination (high-light) conditions, the second row highlights robust detection performance in low-light environments, and the third row presents accurate detection of apples whose color closely resembles surrounding foliage or other fruits.

In Figure 5, first row, apples are observed under strong illumination, where specular reflections and high contrast introduce fragmented visual cues and obscure fruit boundaries in the RGB image. In this scenario, the depth mask highlights foreground structures and enables effective level separation, suppressing background foliage and shadowed regions. As a result, the segmentation stage preserves coherent fruit regions, allowing stable and accurate 2D detection despite intense lighting conditions. The corresponding 3D processing results in Figure 6, first row demonstrate that depth-guided masking supports consistent fruit-level point cloud reconstruction, leading to reliable three-dimensional localisation even when RGB appearance is distorted by illumination effects.

In Figure 5, second row, scenes captured under low-light conditions are presented, where reduced contrast and shadowing significantly weaken RGB-based fruit visibility. Despite these challenges, the proposed framework maintains robust 2D detection by integrating depth-derived cues with instance-level segmentation, preventing missed detections caused by insufficient colour information. As shown in Figure 6, second row, the reconstructed 3D views remain spatially coherent, and fruit-level localisation is preserved, indicating that the proposed depth-aware back-projection effectively compensates for degraded RGB cues in low-illumination environments.

The third row of Figure 5 illustrates cases in which apple colour and appearance closely resemble surrounding foliage or neighbouring fruits, resulting in severe colour ambiguity. In such scenarios, conventional RGB-based methods often struggle to distinguish fruit boundaries. The proposed method overcomes this limitation by exploiting depth-guided masking and instance-level segmentation, enabling accurate separation of fruit regions

from visually similar backgrounds. This improved discrimination is reflected in Figure 6, third row, where distinct fruit-level 3D point sets are reconstructed and precise spatial localisation is achieved despite strong appearance similarity.

Overall, the qualitative results in Figures 5 and 6 demonstrate that the proposed framework achieves reliable 2D apple detection and accurate 3D localisation across a wide range of challenging illumination and appearance conditions. These findings confirm that the tight integration of depth-aware filtering, dual-detector perception, and geometric reasoning significantly enhances robustness in realistic orchard environments, supporting the suitability of the proposed method for deployment in real-world robotic harvesting applications.

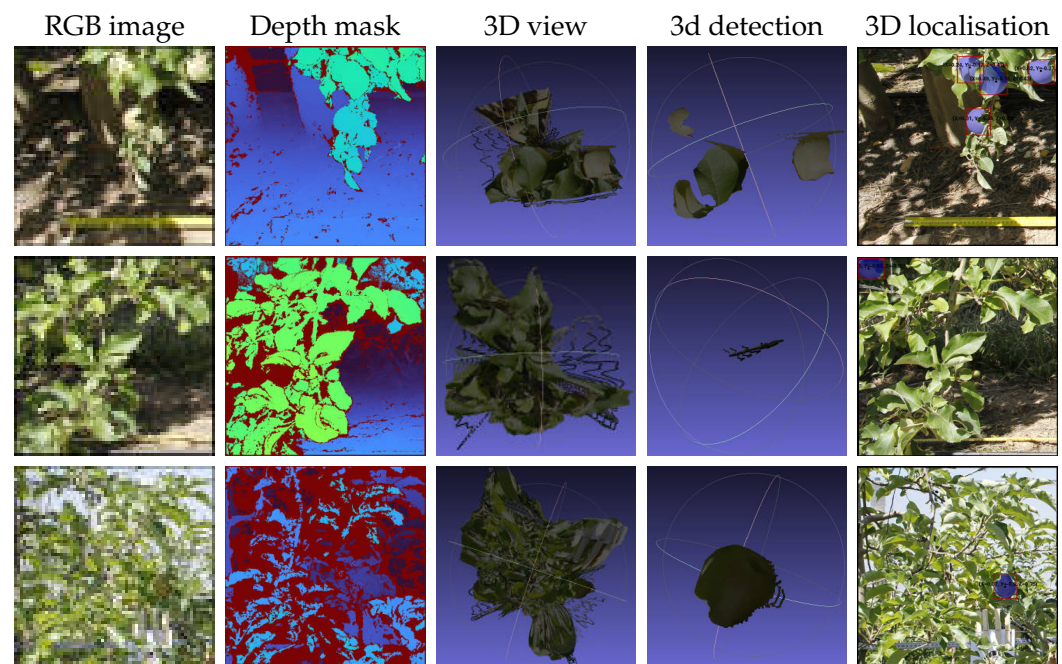


Figure 6. The top row presents 3D localisation results obtained under intense lighting conditions, the middle row illustrates reliable performance in scenes affected by limited illumination, and the bottom row demonstrates successful 3d localisation of apples whose color and appearance are highly similar to surrounding fruits and foliage.

Although the proposed framework achieves robust performance in most orchard scenarios, some failure cases were observed under extreme conditions. Detection errors mainly occur when severe occlusion, extreme illumination, and similar fruit-background colour are present simultaneously, which can significantly degrade both appearance-based features and depth reliability. When these factors co-exist, apples may become indistinguishable from surrounding foliage and background structures, leading to incomplete segmentation or missed detections. These observations indicate that the current method has reduced reliability in highly adverse visual conditions. Future work will explore incorporating temporal information from video sequences, illumination-invariant feature learning, and multi-view perception to further improve robustness under such challenging scenarios.

The comparison methods reported in Table 1 were selected to represent a broad spectrum of widely adopted and state-of-the-art deep learning architectures for object detection and instance segmentation in agricultural and outdoor vision applications. Lightweight single-stage detectors such as MobileNetV3-based models are included to reflect approaches prioritising real-time performance on embedded platforms. Two-stage detectors, including Faster R-CNN and Mask R-CNN, are chosen as strong baselines that emphasise detection accuracy and instance-level segmentation quality. In addition, improved YOLOv8 variants and DaSNet are incorporated as recent high-performance models that have demonstrated

competitive results in fruit detection and related agricultural perception tasks. Together, these methods cover different design philosophies in terms of network complexity, inference speed, and accuracy, enabling a comprehensive and fair evaluation of the proposed framework against current mainstream approaches.

Table 1. Performance comparison of different apple detection methods.

Method	Accuracy (%)	Recall (%)	F1-Score (%)	mAP (%)	FPS
Improved YOLOv8	95.2	87.4	91.0	93.3	72.5
DaSNet	92.1	82.1	86.9	88.4	41.2
MobileNetV3	93.8	79.7	86.1	87.1	90.5
Faster R-CNN	94.0	73.1	82.0	81.2	50.8
Mask R-CNN	96.0	92.7	85.2	89.0	78.6
Proposed Method	98.9	92.8	93.3	94.2	86.42

Table 1 presents a comparison between the proposed method and several representative apple detection approaches using accuracy, recall, F1-score, mean Average Precision (mAP), and inference speed. The proposed method achieves the highest overall detection accuracy (98.9%) and mAP (94.2%), indicating strong capability in both apple classification and localisation under complex orchard conditions.

The proposed method also records the highest F1-score (93.3%), demonstrating a well-balanced trade-off between detection precision and recall. Although Mask R-CNN achieves a comparable recall value, its lower accuracy, F1-score, and inference speed indicate reduced overall robustness. In contrast, the proposed approach maintains high recall (92.8%) while improving detection consistency and localisation reliability, which is particularly important for downstream three-dimensional perception tasks.

From a computational perspective, MobileNetV3 achieves the highest inference speed due to its lightweight architecture; however, this advantage comes at the cost of reduced accuracy and mAP. The proposed method maintains a high inference speed of 86.42 FPS while simultaneously delivering superior detection and localisation performance. This balance highlights its suitability for real-time deployment on embedded platforms.

Overall, while some baseline methods outperform the proposed approach in isolated metrics such as inference speed or recall, none achieve comparable performance across all evaluation criteria. The proposed method consistently provides strong performance in accuracy, F1-score, mAP, and real-time capability, confirming its effectiveness for practical orchard-based apple detection applications.

5. Conclusions

This study proposed a robust RGB-D perception framework for accurate apple detection and three-dimensional localisation in complex orchard environments. By integrating a dual-detector two-dimensional perception strategy based on YOLOv8 and Mask R-CNN with depth-aware back-projection, the proposed method improves detection stability and localisation accuracy under challenging conditions such as heavy foliage occlusion, inter-fruit overlap, illumination variation, and background clutter. The dual-detector fusion leverages complementary bounding-box and instance-level segmentation information to enhance instance separation in densely clustered scenes, while depth-consistent back-projection enables reliable reconstruction of fruit-level three-dimensional point sets and accurate coordinate estimation through robust centroid computation. Extensive experiments on public benchmark datasets and real-world orchard data collected in Lenswood, South Australia, demonstrate that the proposed method achieves improved detection accuracy, balanced precision and recall, and real-time performance on an embedded computing platform. In future work, multi-view perception will be achieved by fusing RGB-D observations from either multiple calibrated cameras or a moving camera mounted on the

robotic manipulator. Extrinsic calibration between cameras and the robot base will enable alignment of all views into a common coordinate frame, followed by geometric fusion of overlapping point clouds to obtain more complete fruit surface reconstructions. These multi-view 3D representations will be used to generate grasp candidates based on surface geometry and visibility. Sampling-based motion planners such as RRT or RRT* will then compute collision-free trajectories for executing the selected grasps. This integration will enable reliable coupling between multi-view perception and robotic grasp planning in cluttered orchard environments.

Author Contributions: Conceptualization, T.H. and P.S.; methodology, T.H.; software, T.H.; validation, T.H., P.S. and L.K.; formal analysis, T.H.; investigation, T.H.; resources, P.S. and L.K.; data curation, T.H.; writing—original draft preparation, T.H.; writing—review and editing, T.H., P.S. and L.K.; visualization, T.H.; supervision, P.S. and L.K.; project administration, P.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The datasets used and/or generated during the current study are not publicly available, as they are directly related to ongoing research activities. However, the data may be obtained from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Zhao, Y.; Gong, L.; Huang, Y.; Liu, C. A Review of Key Techniques of Vision-Based Control for Harvesting Robot. *Comput. Electron. Agric.* **2016**, *127*, 311–323.
2. Indolia, S.; Goswami, A.K.; Mishra, S.P.; Asopa, P. Conceptual Understanding of Convolutional Neural Network—A Deep Learning Approach. *Procedia Comput. Sci.* **2018**, *132*, 679–688.
3. Wei, J.; As’arry, A.; Rezali, K.A.M.; Yusoff, M.Z.M.; Ma, H.; Zhang, K. A Review of YOLO Algorithm and Its Applications in Autonomous Driving Object Detection. *IEEE Access* **2025**, *13*, 93688–93711.
4. Ren, S.; He, K.; Girshick, R.; Sun, J. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *Adv. Neural Inf. Process. Syst.* **2015**, *28*, 91–99.
5. Abdulla, W. Mask R-CNN for Object Detection and Instance Segmentation on Keras and TensorFlow. GitHub Repository. 2017. Available online: https://github.com/matterport/Mask_RCNN (accessed on 1 January 2026).
6. Zhong, H.; Wang, H.; Wu, Z.; Zhang, C.; Zheng, Y.; Tang, T. A Survey of LiDAR and Camera Fusion Enhancement. *Procedia Comput. Sci.* **2021**, *183*, 579–588.
7. Gené-Mola, J.; Llorens, J.; Rosell-Polo, J.R.; Gregorio, E.; Arnó, J.; Solanelles, F.; Martínez-Casasnovas, J.A.; Escolà, A. Assessing the Performance of RGB-D Sensors for 3D Fruit Crop Canopy Characterization under Different Operating and Lighting Conditions. *Sensors* **2020**, *20*, 7072.
8. Yoshida, T.; Kawahara, T.; Fukao, T. Fruit Recognition Method for a Harvesting Robot with RGB-D Cameras. *ROBOMECH J.* **2022**, *9*, 15.
9. Fu, L.; Gao, F.; Wu, J.; Li, R.; Karkee, M.; Zhang, Q. Application of Consumer RGB-D Cameras for Fruit Detection and Localization in Field: A Critical Review. *Comput. Electron. Agric.* **2020**, *177*, 105687.
10. Gao, F.; Fang, W.; Sun, X.; Wu, Z.; Zhao, G.; Li, G.; Li, R.; Fu, L.; Zhang, Q. A Novel Apple Fruit Detection and Counting Methodology Based on Deep Learning and Trunk Tracking in Modern Orchard. *Comput. Electron. Agric.* **2022**, *197*, 107000.
11. Li, T.; Feng, Q.; Qiu, Q.; Xie, F.; Zhao, C. Occluded Apple Fruit Detection and Localization with a Frustum-Based Point-Cloud-Processing Approach for Robotic Harvesting. *Remote Sens.* **2022**, *14*, 482.
12. Wang, Z.; Xu, J.; Zhang, X.; Shen, A.; Zhang, Z.; Xun, Y. In Situ Detection and Measurement of Broccoli Heads under Different Lighting Conditions Using Proximal Remote Sensing. *J. Field Robot.* **2025**, *42*, 4101–4114.
13. Zhao, Z.; Hicks, Y.; Sun, X.; Luo, C. FruitQuery: A Lightweight Query-Based Instance Segmentation Model for In-Field Fruit Ripeness Determination. *Smart Agric. Technol.* **2025**, *12*, 101068. <https://doi.org/10.1016/j.atech.2025.101068>.
14. Ma, F.; Cavalheiro, G.V.; Karaman, S. Self-Supervised Sparse-to-Dense: Self-Supervised Depth Completion from LiDAR and Monocular Camera. *Proc. IEEE Int. Conf. Robot. Autom.* **2019**, 3288–3295.

15. Zhou, Y.; Yu, C.; Zhang, J.; Liu, Y.; Kan, J.; Zou, X.; Zhang, K.; Li, H.; Zhang, S.; Wu, F. Vision-Based Perception and Execution Decision-Making for Fruit Picking Robots Using Generative AI Models. *Machines* **2024**, *14*, 117. <https://doi.org/10.3390/machines14010117>.
16. Myint, B.B.; Onizuka, T.; Tin, P.; Aikawa, M.; Kobayashi, I.; Zin, T.T. Automated Cattle Detection Using Mask R-CNN and IoU-Based Tracking with a Single Side-View Camera. *Int. J. Innov. Comput. Inf. Control* **2024**, *20*, 1439–1447.
17. Molina, J.M.; Llerena, J.P.; Usero, L.; Patricio, M.A. Advances in Instance Segmentation: Technologies, Metrics and Applications in Computer Vision. *Neurocomputing* **2025**, *587*, 129584.
18. Mihajlovic, M.; Marjanovic, M. Enhancing Instance Segmentation in High-Resolution Images Using Slicing-Aided Hyper Inference and Spatial Mask Merging Optimized via R-Tree Indexing. *Mathematics* **2025**, *13*, 3079.
19. Sharma, R.; Saqib, M.; Lin, C.T.; Blumenstein, M. A Survey on Object Instance Segmentation. *SN Comput. Sci.* **2022**, *3*, 499.
20. Valada, A.; Mohan, R.; Burgard, W. Self-Supervised Model Adaptation for Multimodal Semantic Segmentation. *Int. J. Comput. Vis.* **2020**, *128*, 1239–1285.
21. Liu, Y.; Matsui, K.; Kageyama, Y.; Shirai, H.; Ishizawa, C. Improving a Faster R-CNN Model for Vehicle Detection and Human Action Recognition at Night via Infrared Thermal Imaging Using Transfer Learning. *Int. J. Innov. Comput. Inf. Control* **2024**, *20*, 1573–1585.
22. Park, K.; Kim, S.; Sohn, K. High-Precision Depth Estimation Using Uncalibrated LiDAR and Stereo Fusion. *IEEE Trans. Intell. Transp. Syst.* **2020**, *21*, 321–335.
23. Mousavian, A.; Anguelov, D.; Flynn, J.; Kosecka, J. 3D Bounding Box Estimation Using Deep Learning and Geometry. *Comput. Vis. Pattern Recognit.* **2017**, 5632–5640.
24. Lu, S.; Chen, W.; Zhang, X.; Karkee, M. Canopy-Attention-YOLOv4-Based Immature/Mature Apple Fruit Detection on Dense-Foliage Tree Architectures for Early Crop Load Estimation. *Comput. Electron. Agric.* **2022**, *193*, 106696.
25. Wang, Y.; Chen, Y. Fruit Morphological Measurement Based on Three-Dimensional Reconstruction. *Agronomy* **2020**, *10*, 455.
26. Li, Z.; Xu, X.; Xu, T.; Gao, Y.; Rao, X. Process, Challenges and Solutions of Fruit 3D Reconstruction: A Review. *Comput. Electron. Agric.* **2025**, *239*, 111068.
27. Ge, Y.; Xiong, Y.; From, P.J. Three-Dimensional Location Methods for the Vision System of Strawberry-Harvesting Robots: Development and Comparison. *Precis. Agric.* **2023**, *24*, 764–782.
28. He, X.; Zhang, X.; Wang, Y.; Ji, H.; Duan, X.; Guo, F. Spatial Attention Frustum: A 3D Object Detection Method Focusing on Occluded Objects. *Sensors* **2022**, *22*, 2366.
29. Wang, H.; Zhang, G.; Cao, H.; Hu, K.; Wang, Q.; Deng, Y.; Gao, J.; Tang, Y. Geometry-Aware 3D Point Cloud Learning for Precise Cutting-Point Detection in Unstructured Field Environments. *J. Field Robot.* **2025**, *42*, 3063–3076. <https://doi.org/10.1002/rob.22567>.
30. Scutelnic, D.; Daffara, C.; Muraddore, R.; Weinmann, M.; Jutzi, B. Multi-Model Ensembles for Object Detection in Multispectral Images: A Case Study for Precision Agriculture. *Comput. Electron. Agric.* **2026**, *240*, 111213. <https://doi.org/10.1016/j.compag.2025.111213>.
31. Almahasneh, M.; Paiement, A.; Xie, X.; Aboudarham, J. MLMT-CNN for Object Detection and Segmentation in Multi-Layer and Multi-Spectral Images. *Mach. Vis. Appl.* **2022**, *33*, 9. <https://doi.org/10.1007/s00138-021-01261-y>.
32. Kutryev, A.; Andriyanov, N.; Khort, D.; Smirnov, I.; Zubina, V. Adaptive CNN Ensemble for Apple Detection: Enabling Sustainable Monitoring Orchard. *AgriEngineering* **2025**, *7*, 369.
33. Yang, X.; Gao, Y.; Yin, M.; Li, H. Automatic Apple Detection and Counting with AD-YOLO and MR-SORT. *Sensors* **2024**, *24*, 7012.
34. Khosravi, H.; Saedi, S.I.; Rezaei, M. Real-Time Recognition of on-Branch Olive Ripening Stages by a Deep Convolutional Neural Network. *Sci. Hortic.* **2021**, *287*, 110252.
35. Wu, H.; Chen, K.; Ni, M.; Ma, L. Strip Surface Defect Detection Based on Improved YOLOv7. *Int. J. Innov. Comput. Inf. Control* **2024**, *20*, 1493–1507.
36. Zuo, F.; Liu, J.; Chen, Z.; Zhang, H.; Fu, M.; Wang, L. Multilevel Fine-Grained Features-Based General Framework for Object Detection. *IEEE Trans. Cybern.* **2024**, *54*, 6921–6933.
37. Jaw, D.-W.; Huang, S.-C.; Lu, Z.-H.; Fung, B.C.M.; Kuo, S.-Y. Multidomain Object Detection Framework Using Feature Domain Knowledge Distillation. *IEEE Trans. Cybern.* **2024**, *54*, 4643–4651.
38. Yan, B.; Li, X. RGB-D Camera and Fractal-Geometry-Based Maximum Diameter Estimation Method of Apples for Robot Intelligent Selective Graded Harvesting. *Fractal Fract.* **2024**, *8*, 649.
39. Kang, H.; Zhou, H.; Chen, C. Visual Perception and Modeling for Autonomous Apple Harvesting. *IEEE Access* **2020**, *8*, 62151–62163.
40. Sekharamanthy, P.K.; Melgani, F.; Malacarne, J.; Ricci, R.; Almeida, D., S., R.; Marcato, J. A Seamless Deep Learning Approach for Apple Detection, Depth Estimation, and Tracking Using YOLO Models Enhanced by Multi-Head Attention Mechanism. *Computers* **2024**, *13*, 83.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.